

**CQ SOIL
TESTING**



AS2870 Site Classification

SITE ADDRESS:

Proposed New Office Building.
Lot 44 (SP140243)
Stanwell Power Station.
Power Station Road, Stanwell.

Prepared for:

Dezign Group QLD Pty Ltd
T.A Dezign Elements.

Job Number:

CQ29760

Issue Date:

19/12/2025



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Client & Document Information

Client: Deziqn Elements
Project: Lot 44 (SP140243)
Power Station Road, Stanwell

Investigation Type: **Site Classification**
Job Number: CQ29760
Date of Issue: 19/12/2025

Contact Information

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Document Control

Version	Date	Author	Design Drawings	Reviewer	Reviewer Initials
A	19/12/2025	D Pinkney	N/A	P Munro	PM

QBCC Subsidence Policy

In accordance with the QBCC “Queensland Building and Construction Commission” the contractor must supply the site classifier with the information in Table 1. The contractor, or the contractor representative (CR), may require the site classifier (SC) gather all or part of this information and the SC must satisfy themselves that all of the “relevant” information has been considered.

If all of the information listed below is not supplied by the contractor or the contractor does not wish the SC to recover said information (at cost) the contractor may be in breach of the no fault provisions of the QBCC’s Policy for Rectification of Building Work and may be held responsible for subsidence or settlement of a building.

Table 1

Element	Supplied/ Considered	Remarks
Property description and site address	✓	Supplied by Client
Plan and/or survey	✓	Supplied by Client
Contour of the site	✗	Nil Supplied
Location of trees, vegetation etc identified	✓	Identified by SC
Location and identification of potential overland flow	✓	Identified by SC
The footprint of proposed building and platform levels	✓	Supplied by Client
Location of proposed or existing cut and fill	✗	Nil Supplied
Appropriate land searches	✗	Nil Supplied

The following (Table 2) is a summary of the information required under the QBCC relating specifically to the SC. Information supplied in this summary is to be read in conjunction with the entire report attached. All relevant data used to ascertain the classification is documented in the report.

Table 2

Element	Remarks
Total number of excavations	3
Minimum of two excavations in building footprint	✓
Soil samples recovered	Undisturbed
Laboratory test performed	Shrink/Swell
Predicted Surface Movement	31 - 40 mm
Expected movement potential for “P” sites in the absence of uncontrolled fill and soft soil	NA

1.0 Introduction

The purpose of this report is to classify the subject allotment in accordance with Australian Standard 2870 Residential Slabs and Footings". From this classification a footing system can be recommended by an experienced/qualified engineer (designer) to suit the proposed structure. This design shall provide adequate performance of the footings under the soil conditions determined at the site.

This site investigation has been carried out by an experienced/qualified soils technician and in accordance with AS 2870. CQ Soil Testing is licensed with Building Services Australia to "Classify Sites".

This report relates exclusively to the proposed new office building at the address stated on page one of this report and has been prepared for the express purpose stated above. This document does not cover any other elements related to construction on the site.

2.0 Site Description

The subject site is a power station which fronts a sealed road.

The proposed construction site is sparsely grassed and there is no evidence of any large trees having been removed from the proposed construction footprint (see attached photographs). The proposed construction site is essentially level and is considered to have poor drainage. Surface water will drain toward the east. Surface water from the adjoining allotments may traverse the site. A site sketch is attached to this report.

There is no evidence of fill having been placed onto the proposed construction site.

3.0 Soil Profile

Boreholes carried out at the site (refer attached site sketch for approximate localities) indicate a soil profile of silty clay which is underlain by silty sandy clay then silty clay then residual soil (see detailed logs). Tungsten carbide drill bit refusal was not encountered. Laboratory testing was carried out on typical soil sample/s to assess the potential of the underlying soils to exhibit shrink/swell characteristics and any underlying moisture conditions. Details of the laboratory test results are contained in Section 4.

- Groundwater was not encountered during the site investigation.
- Residual soil was encountered during the site investigation.

It is possible that the soil profile may vary across the site from those shown in the bore logs which were used for this site classification. CQ Soil Testing are required to be notified if different conditions are encountered during construction. No allowance has been made for any substantial earthworks on the site or importing building platform material. ***The classification provided is based on the borehole, which has the highest characteristic surface movement.***

4.0 SITE CLASSIFICATION AND TARGET STRATA

Based on the findings of the site investigation and subsequent laboratory testing, the predicted surface movement for this site is between 31 - 40 mm which would give a classification of

CLASS "M" **(Moderately Reactive)**

in accordance with Australian Standard 2870, Residential Slabs and Footings. The above classification has not allowed for the possibility of differential surface movement as a result of differing soil types throughout the site or as a result of construction activities. It is the responsibility of the engineer to allow for this possibility in the footing design.

An indicative bearing capacity of greater than 100 kpa was encountered throughout the strata at all borehole locations. Any fill placed over the existing ground shall be pierced through into the existing suitable material. Further note that the placement of reactive material as fill or cutting of the site may change the site's classification.

Where trees exist/ed CQ Soil Testing recommends an experienced arborist be commissioned to quantify the existing size, location, predicted maximum height and type of all relevant trees to aid in the design process. It is the responsibility of the designing engineer to apply the principals of AS2870-2011 Appendix H "Guide to Design of Footings for Trees". The classification herein excludes the effect of trees on the site.

It is noteworthy that soil samples recovered from this site may be tested further to aid in the preparation of a database of Central Queensland soils currently being compiled by CQ Soil Testing. The aim of this database is to further understand the types of soils in the region and their mechanical properties.

If you should have any queries regarding this report, please do not hesitate to contact the undersigned at your convenience.

Yours faithfully



SCOTT WALTON
Laboratory Manager

Site/Soil Characteristics and Classification

A. Classification by characteristic surface movement as per AS2870-2011

Site Classification Symbols	Y's Range Value	Generalised Description (Guide Only)
'S'	0 – 20 mm	Slightly reactive clay sites which may experience only slight ground movement due to moisture changes
'M'	21 – 40 mm	Moderately reactive clay or silt sites which may experience moderate ground movement due to moisture changes
'H1'	41 – 60 mm	Highly reactive clay sites which may experience high ground movement due to moisture changes
'H2'	61 – 75 mm	Highly reactive clay sites which may experience very high ground movement due to moisture changes
'E'	>75 mm	Extremely reactive clay sites which may experience extreme ground movement due to moisture changes
'P'	N/A	Problem sites which generally have soils associated with uncontrolled fill, abnormal moisture conditions (trees), soft or collapsing soils, landslip etc...

B. Laboratory Test Results

Borehole Location	3	Borehole Location		Borehole Location	
Depth Range of Sample (m)	0.5–1.2	Depth Range of Sample (m)		Depth Range of Sample (m)	
Natural MC %	13	Natural MC %		Natural MC %	
% Passing 75 um Sieve	ND	% Passing 75 um Sieve		% Passing 75 um Sieve	
Liquid Limit %	ND	Liquid Limit %		Liquid Limit %	
Plastic Index %	ND	Plastic Index %		Plastic Index %	
Linear Shrinkage %	ND	Linear Shrinkage %		Linear Shrinkage %	
Shrink Swell Index	1.8	Shrink Swell Index		Shrink Swell Index	
Pocket Penetrometer kPa	ND	Pocket Penetrometer kPa		Pocket Penetrometer kPa	

C. Permeability Test Results AS1547-2012

Test Hole Number	Depth Of Test Hole	Range Tested	Permeability M/Day
NA	500 mm	250 – 500 mm	NA

Soil Logs



BOREHOLE 1			
Depth (m)	Visual Class'n Symbol	Visual Description of Material	
0.0	CI	Silty CLAY, medium plasticity, brown/light brown mottled, D, ST.	
0.5			
0.5	CI	Silty Sandy CLAY, medium plasticity, fine to medium grained, trace of fine to medium gravel, brown, D, VST-H w/depth.	
1.5			
1.5	CI/RS	Silty CLAY, medium plasticity, trace of fine to medium grained, brown/dark brown mottled greyish brown w/depth, D, H.	
3.0			
Residual soil.			
Borehole terminated at 3.0 m			

DCP TEST RESULTS		
Depth (mm)	Blows per 100 mm	Indicative kPa
100	4	120
200	3	100
300	4	120
400	4	120
500	5	160
600	7	200
700	9	250
800	8	200
900	9	250
1000	10	250
1100	12	250
1200	12	250
1300	>15	>300
1400	Drill	-
1500	Drill	-
1600	>15	>300
1700	Drill	-
1800	Drill	-
1900	Drill	-
2000	Drill	-
2100	>15	>300
2200		
2300		
2400		
2500		
2600		
2700		
2800		
2900		
3000		
3100		
3200		
3300		
3400		
3500		
3600		
3700		
3800		
3900		
4000		

Soil Logs



BOREHOLE 2			
Depth (m)	Visual Class'n Symbol	Visual Description of Material	
0.0	CI	<u>Silty CLAY</u> , medium plasticity, brown/light brown mottled, D, ST.	
0.5			
0.5	CI	<u>Silty Sandy CLAY</u> , medium plasticity, fine to medium grained, trace of fine to medium gravel, brown, D, VST-H w/depth.	
2.0			
2.0	CI/RS	<u>Silty CLAY</u> , medium plasticity, trace of fine to medium grained, brown/dark brown mottled, D, H.	
3.0			
Residual soil.			
Borehole terminated at 3.0 m			

DCP TEST RESULTS		
Depth (mm)	Blows per 100 mm	Indicative kPa
100	3	100
200	3	100
300	4	120
400	4	120
500	4	120
600	7	200
700	8	200
800	8	200
900	9	250
1000	10	250
1100	12	250
1200	13	300
1300	>15	>300
1400	Drill	-
1500	Drill	-
1600	Drill	-
1700	Drill	-
1800	Drill	-
1900	Drill	-
2000	Drill	-
2100	14	300
2200	14	300
2300	>15	>300
2400		
2500		
2600		
2700		
2800		
2900		
3000		
3100		
3200		
3300		
3400		
3500		
3600		
3700		
3800		
3900		
4000		

Soil Logs



BOREHOLE 3				
Depth (m)	Visual Class'n Symbol	Visual Description of Material		
0.0	CI	Silty CLAY, medium plasticity, brown/light brown mottled, D, ST.		
0.5				
0.5	CI	Silty Sandy CLAY, medium plasticity, fine to medium grained, trace of fine to medium gravel, brown, D, VST-H w/depth.		
2.5				
2.5	CI/RS	Silty CLAY, medium plasticity, trace of fine to medium grained, brown/dark brown mottled, D, H.		
3.0		Residual soil.		
Borehole terminated at 3.0 m				
MOISTURE CONDITION		CONSISTENCY	RELATIVE DENSITY	Allowable Bearing Pressure calculated using the guidelines in "Determination of Allowable Bearing Pressure under Small Structures" by MI Stockwell (NZ Engineering June 1997)
D – Dry		VS – Very Soft	VL – Very Loose	
M – Moist		S – Soft	L – Loose	
W – Wet		F – Firm	MD – Med Dense	DCP test results are to be used as a guide only to relative density and consistency of soils. Changes in moisture contents or the presence of coarse-grained material can greatly influence the outcome of this test.
		ST – Stiff	D – Dense	
		VST – Very Stiff	VD – Very Dense	
		H – Hard		

DCP TEST RESULTS		
Depth (mm)	Blows per 100 mm	Indicative kPa
100	3	100
200	3	100
300	4	120
400	5	160
500	4	120
600	7	200
700	8	200
800	9	250
900	12	250
1000	>15	>300
1100	Drill	-
1200	Drill	-
1300	Drill	-
1400	Drill	-
1500	Drill	-
1600	11	250
1700	12	250
1800	13	300
1900	>15	>300
2000	Drill	-
2100	Drill	-
2200	Drill	-
2300	Drill	-
2400	Drill	-
2500	Drill	-
2600	13	300
2700	>15	>300
2800		
2900		
3000		
3100		
3200		
3300		
3400		
3500		
3600		
3700		
3800		
3900		
4000		

Photographs

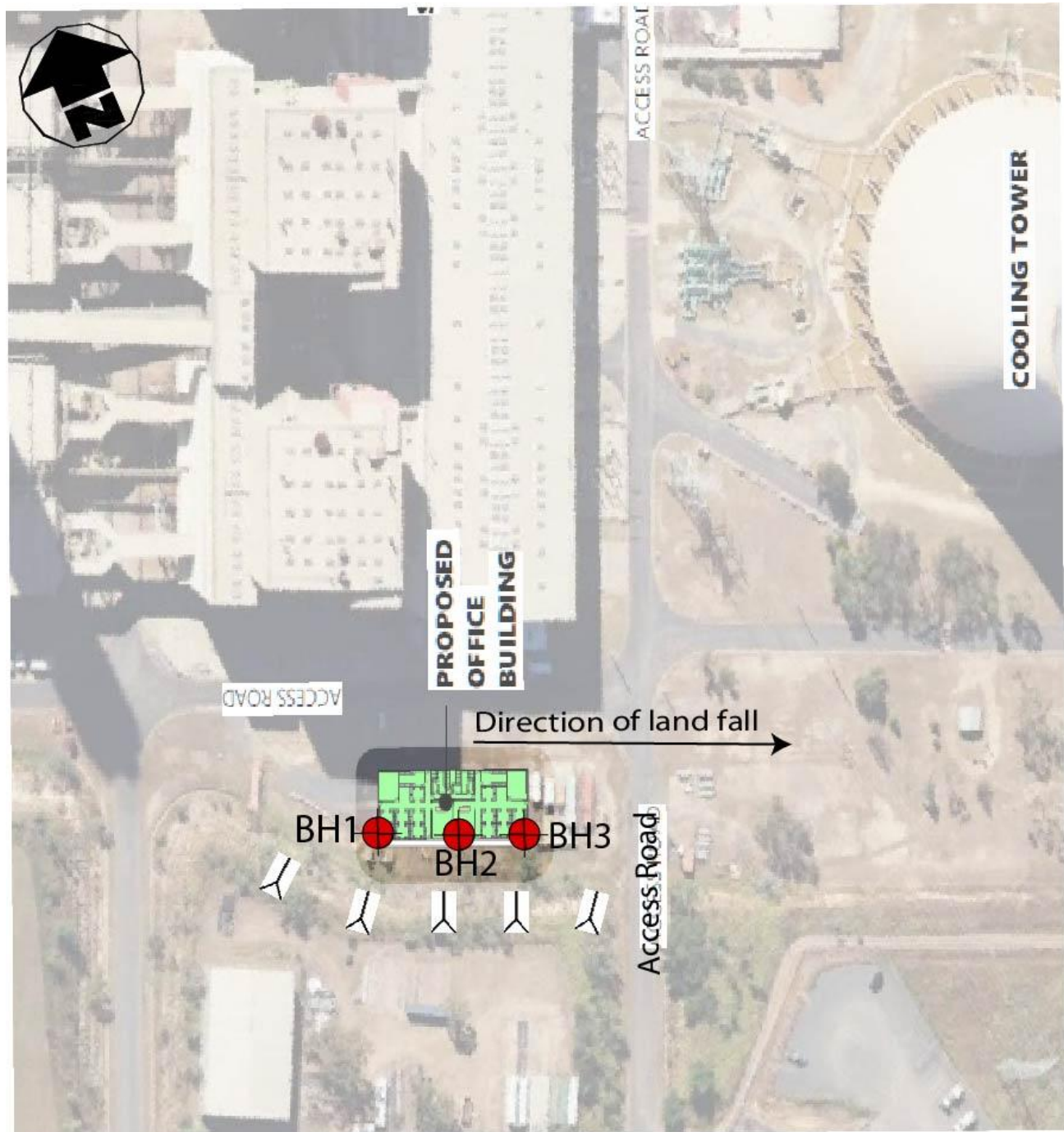


Image 1 – Proposed construction site



Image 2 – Proposed construction site

Site Plan



- Not to scale
- All measurements are to be used as a guide only

LIMITATIONS

1. Recommendations given in this report are based on the information supplied by the client regarding the proposed building construction in conjunction with the findings of the investigation. Any change in construction type, building location or omission in the client supplied information, may require additional testing and/or make the recommendations invalid.
2. The recommendations herein may identify a target soil stratum into which the footings should be founded. The target stratum has been located by the depth in mm of the target stratum's upper horizon boundary below the existing ground surface level at the time of the site investigation. Any cutting or filling works and any surface erosion or deposits subsequent to the site investigation, will alter the measured location of the stratum relative to the surface. Where required, the author should be notified in such cases to confirm the location of the target stratum.
3. The description of the soil given in Section 3.0 of this report is intended as a brief overview of the soil's primary constituents. For a detailed classification of the soil, the reader should refer to the Soil Profile Reports and/or Borehole Reports.
4. Every reasonable effort has been made to locate the test sites so that the borehole profiles are representative of the soil conditions within the area investigated. The client should be made aware, however, that exploration is limited by time available and economic restraints. In some cases, soil conditions can change dramatically over short distances, therefore, even careful exploration programs may not locate all the variations.
5. If soil conditions different from those shown in this report are encountered or are inferred from other sources, then the author must be notified immediately.
6. This report may not be reproduced except in full, and only then with the permission of the entity trading as CQ Soil Testing. The information and site sketch shall only be used and will only be applicable for the development shown on the client-supplied information provided for this site.
7. All information contained within this report is the intellectual property of the entity trading as CQ Soil Testing. All information contained within can only be used for the express purposes of the commissioned scope of works.
8. Any dimensions, contours, slope directions and magnitudes shown on the site sketch plan shall not be used for any building construction or costing calculations. The purpose of the plan is to show the approximate location of field tests only.
9. Any changes made to these recommendations by persons unauthorized by the author will legally be interpreted by that person assuming the responsibility for the long-term performance of the footing system.
10. The recommendations contained in this report have not taken into consideration the long-term effects of any previous, current, or potential subsurface work by mining companies or potential slope instability problems. At the time of writing this report neither our client (nor his agent) nor the local authority had made the author aware that these problems may be affecting this allotment. If a mining subsidence or slope stability assessment is required for this allotment, the recommendations of a suitably qualified geotechnical engineer should be sought.
11. Removal of trees from a site before an investigation can cause significant swelling of the soil over large areas. The removal of large trees from a construction site during development is rarely picked up during the investigation phase and is generally outside the scope of AS2870. Sites affected by large trees are often classified "P". If, during the footing excavation, it is noticed that there are soils with varying moisture contents or evidence of large trees having been removed CQ Soil Testing should be notified immediately.
12. The following documents are available from the CSIRO and QBCC and shall be read and adhered to in relation to this site:
 - Builder's Guide to Preventing Damage to Dwellings- Part 1 Site Investigation and Preparation
<http://www.publish.csiro.au/nid/22/pid/3621.html>
 - Builder's Guide to Preventing Damage to Dwellings- Part 2 Sound Construction Methods
<http://www.publish.csiro.au/nid/22/pid/3661.html>
 - QBCC Subsidence Fact Sheet
<https://www.qbcc.qld.gov.au/sites/default/files/Homeowner%27s%20Guide%20to%20Subsidence.pdf>